

AYUSH KHARE

Machine Design Engineer — FEA Engineer — R&D Engineer

✉ ayushkhare3.14@gmail.com — 📞 +91-9926207572 —  linkedin —  Portfolio Website

Professional Summary

Mechanical Engineer with M.Tech specialization in Design & Manufacturing from IIT Tirupati, experienced in engineering calculations, finite element analysis (FEA), structural validation, machine design, and simulation-driven product development. Skilled in ANSYS Workbench, CATIA V5, SolidWorks, and MATLAB for stress analysis, design optimization, prototype validation, and industrial automation applications.

Demonstrated experience in defense systems, UAV platforms, semiconductor equipment, industrial automation, and machine vision systems. Strong exposure to R&D, cost optimization, design-for-manufacturing (DFM), and cross-functional engineering collaboration for high-performance mechanical systems.

Education

Degree	Institute	Year	CGPA/%
M.Tech – Design & Manufacturing	Indian Institute of Technology Tirupati	2024	7.81
B.Tech – Mechanical Engineering	Gyan Ganga Institute of Technology and Sciences	2021	8.95
Class XII	Nachiketa S.S School	2017	80 %
Class X	Nachiketa S.S School	2015	8.4

Technical Skills

CAD Software	CATIA V5, SolidWorks, AutoCAD
FEA & Simulation	ANSYS Workbench, MATLAB, Static Structural Analysis, Stress Analysis, Structural Validation, Design Optimization
Engineering Skills	Machine Design, Product Design, Engineering Calculations, DFMEA, DFx, QFD, Material Selection, Tolerance Analysis
Automation & Vision Systems	Machine Vision Systems, Industrial Automation, Vision Cameras, Process Optimization
Manufacturing	Prototype Development, Composite Manufacturing, Vacuum Infusion Process (VIP), Assembly Validation
UAV & Control Systems	ArduPilot, RTK GPS Integration, Autonomous Systems
Professional Skills	Project Leadership, Cross-functional Collaboration, Technical Problem Solving, R&D, Cost Optimization

Professional Experience

Design & R&D Engineer April 2025 – Present
Nordic Design House

- Responsible for product design, engineering calculations, structural validation, simulation, and R&D activities for industrial automation and mechanical systems.
- Worked on innovative concept development, manufacturability improvement, cost optimization, and simulation-driven product development.
- Performed hand calculations and Finite Element Analysis (FEA) using ANSYS Workbench for stress analysis, deformation analysis, and factor of safety validation.
- Collaborated with manufacturing, automation, and quality teams for process optimization and zero-defect production systems.

Key Projects

- **Spring Dowel Prototype Development**
 - Led R&D and prototype development for a spring dowel machine previously dependent on imported systems.
 - Performed engineering calculations, structural simulations, and ANSYS validation for prototype development.
 - Optimized design for manufacturability, structural reliability, and cost reduction.
- **Rocker Arm Assembly Automation**
 - Worked as Project Leader for automation of rocker arm assembly process for critical engine components.
 - Developed zero-defect assembly concepts integrating machine vision cameras and automated inspection systems.
 - Improved process repeatability, productivity, and quality assurance through automation strategies.
- **SKF Socket Structural Analysis for BDS Fasteners**
 - Performed detailed engineering calculations and ANSYS structural simulations for SKF socket validation.

- Evaluated stress concentration, deformation behavior, and load-bearing capability under operating conditions.
- Supported structural reliability assessment and design improvement initiatives.

- **Vision Inspection System – Naminath**

- Led development activities for machine vision-based fastener inspection systems.
- Worked on defect detection logic, inspection methodologies, and intelligent automation concepts.
- Contributed to innovative quality control and industrial automation solutions.

Design Engineer

July 2024 – April 2025

Transvahan Technologies India Pvt. Ltd., Bengaluru

- Performed engineering calculations and structural validation for products designed for Indian Army applications.
- Conducted static structural analysis and optimization using ANSYS Workbench to improve product reliability and reduce structural weight.
- Optimized components for manufacturability and cost reduction while maintaining structural integrity and safety standards.
- Collaborated with design and manufacturing teams during prototype development and validation activities.

Intern – Design Engineer

March 2023 – June 2024

Prima Innotech LLC, USA

- Designed and optimized CAD models for prototype exfoliation and Ni electroplating apparatus for semiconductor applications.
- Applied DfX methodologies, QFD, material selection, and cost optimization techniques during product development.
- Conducted structural analysis and validation using ANSYS Workbench.
- Supported manufacturing and assembly activities at IIT Tirupati facilities.
- Developed conceptual design of wafer exfoliation systems integrating controlled spalling techniques inspired by IBM research.
- Designed traction model for an autonomous electric vehicle in MATLAB Simulink to evaluate motor torque, acceleration, battery consumption, and vehicle performance.

Major Projects

Design and Development of Fixed Wing Drone

- Designed autonomous fixed-wing drone targeting long-endurance flight operations.
- Performed structural analysis and lightweight composite airframe design.
- Worked on carbon fiber composite manufacturing using Vacuum Infusion Process (VIP).
- Integrated RTK GPS and ArduPilot for autonomous takeoff and landing operations.

Drone Launcher System

- Designed and developed indigenous drone launcher capable of launching drones up to 20 kg payload capacity.
- Performed CAD modeling, structural analysis, optimization, and fabrication activities.
- Developed cost-effective launcher architecture for UAV and defense applications.

Electric Kart

- Designed CAD model using CATIA V5 for National Electric Kart Championship.
- Conducted structural analysis using ANSYS Workbench for chassis validation and safety assessment.

Achievements

- Awarded Chancellor Scholarship by RGPV University.
- Selected for Smart India Hackathon 2018 under Security and Surveillance domain.
- Led multiple R&D and industrial automation projects involving structural simulation and machine vision systems.

Positions of Responsibility

Placement Representative – IIT Tirupati

- Coordinated placement activities for Design & Manufacturing branch.

PFAM International Conference Coordinator – IIT Tirupati

- Managed guest coordination, technical sessions, and conference scheduling activities.

Extra-Curricular Activities

- Completed NCC 'B' and NCC 'C' Certificates.
- Represented hostel volleyball team in inter-hostel competitions.

Areas of Interest

Finite Element Analysis (FEA), Structural Simulation, Machine Design, Product Development, Industrial Automation, UAV Systems, Defense Technology, Manufacturing Optimization, Mechanical R&D